

30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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Programming Language	Python

Assessment

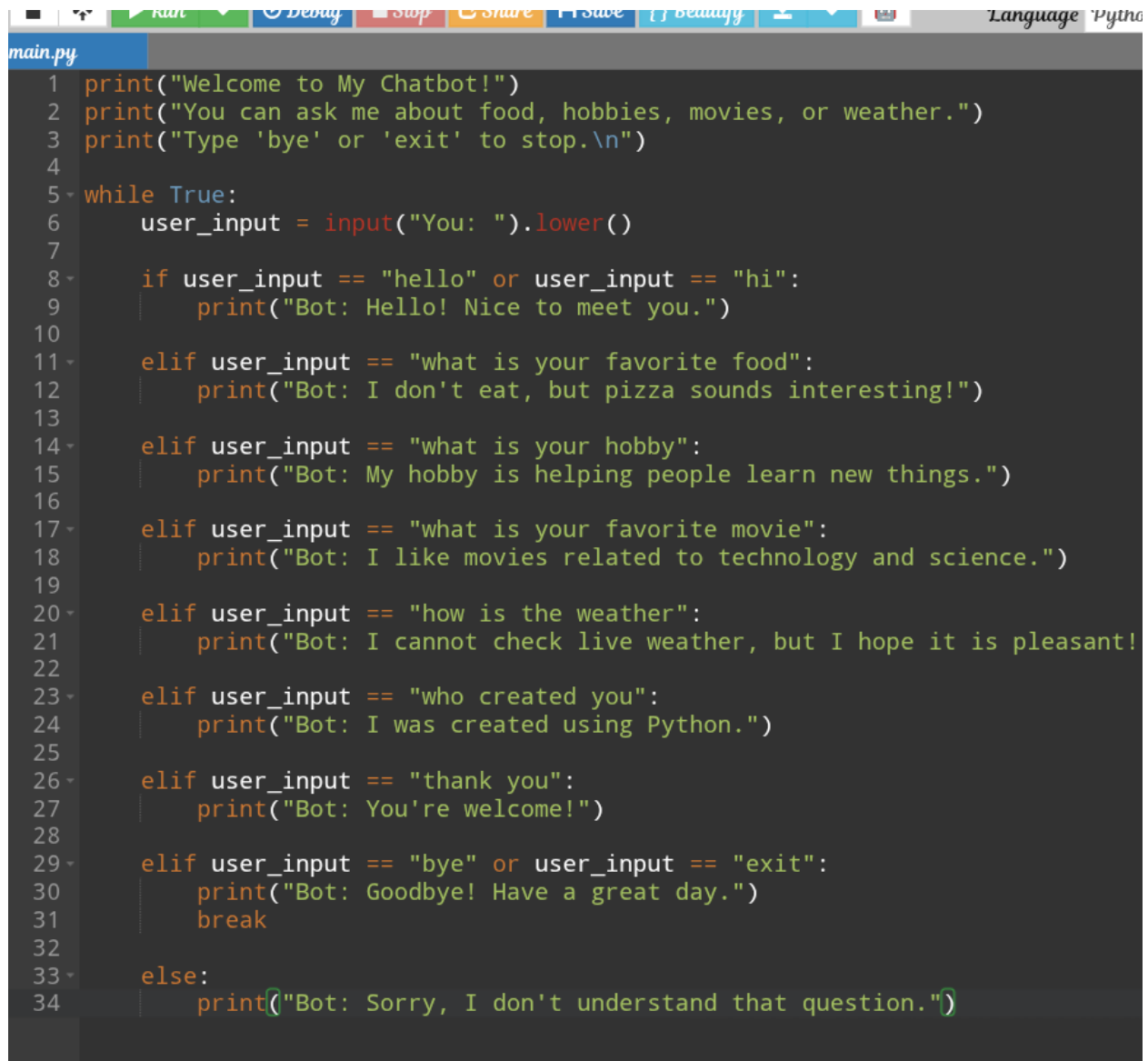
A simple rule-based conversational chatbot was developed in Python. The chatbot accepts user input continuously, recognizes greetings and different questions, responds accordingly, and exits when the user types bye or exit.

Demonstration Evidence

The following pages contain screenshots of the Python implementation and the terminal execution.

Python Chatbot Implementation

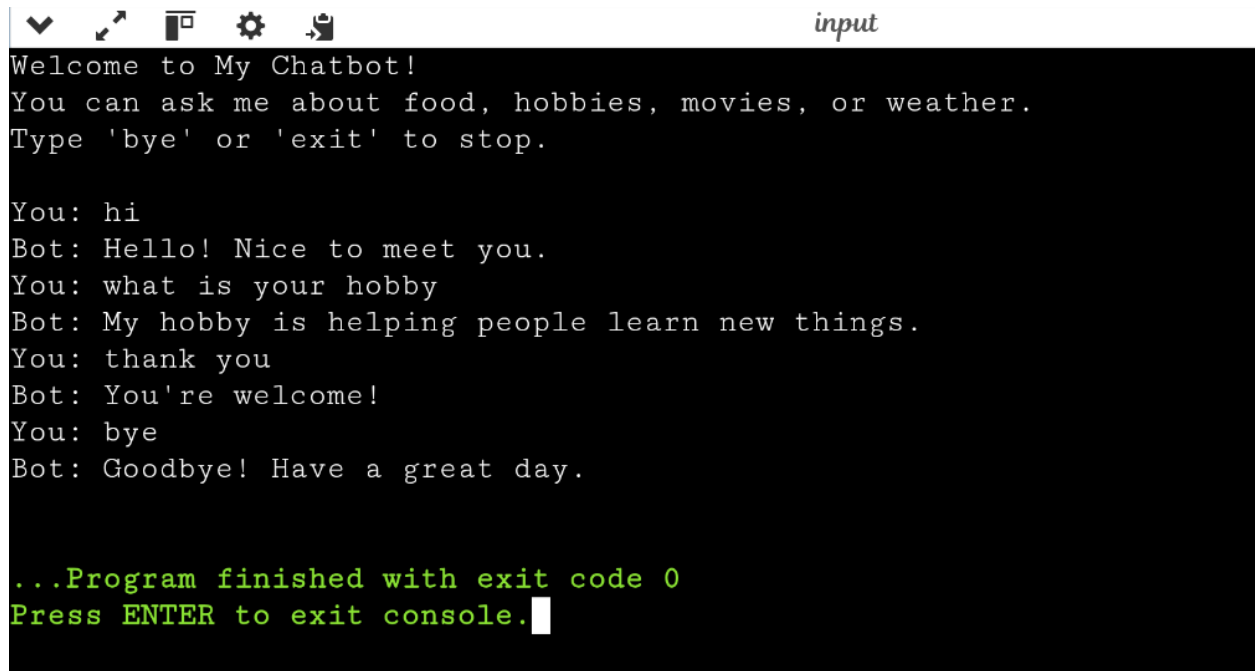
Screenshot of the Python program implementation.

A screenshot of a Python IDE window titled 'main.py'. The code implements a simple chatbot with a while loop that prompts the user for input. The chatbot has several predefined responses based on the user's input. The code is as follows:

```
1 print("Welcome to My Chatbot!")
2 print("You can ask me about food, hobbies, movies, or weather.")
3 print("Type 'bye' or 'exit' to stop.\n")
4
5 while True:
6     user_input = input("You: ").lower()
7
8     if user_input == "hello" or user_input == "hi":
9         print("Bot: Hello! Nice to meet you.")
10
11    elif user_input == "what is your favorite food":
12        print("Bot: I don't eat, but pizza sounds interesting!")
13
14    elif user_input == "what is your hobby":
15        print("Bot: My hobby is helping people learn new things.")
16
17    elif user_input == "what is your favorite movie":
18        print("Bot: I like movies related to technology and science.")
19
20    elif user_input == "how is the weather":
21        print("Bot: I cannot check live weather, but I hope it is pleasant!")
22
23    elif user_input == "who created you":
24        print("Bot: I was created using Python.")
25
26    elif user_input == "thank you":
27        print("Bot: You're welcome!")
28
29    elif user_input == "bye" or user_input == "exit":
30        print("Bot: Goodbye! Have a great day.")
31        break
32
33    else:
34        print("Bot: Sorry, I don't understand that question.")
```

Chatbot Execution and Test Output

Terminal output showing greeting, question handling, an unknown message, and the exit command.

A terminal window with a dark background and light green text. The window has a title bar with standard Linux window controls (minimize, maximize, close) and a gear icon for settings. The text inside the terminal shows a chatbot's welcome message, followed by a series of user inputs and bot responses. The interaction ends with a message indicating the program finished successfully with exit code 0.

```
input
Welcome to My Chatbot!
You can ask me about food, hobbies, movies, or weather.
Type 'bye' or 'exit' to stop.

You: hi
Bot: Hello! Nice to meet you.
You: what is your hobby
Bot: My hobby is helping people learn new things.
You: thank you
Bot: You're welcome!
You: bye
Bot: Goodbye! Have a great day.

...Program finished with exit code 0
Press ENTER to exit console.
```